

RENAL PHYSIOLOGY

A Comprehensive Study Guide

Compiled by the EverythingABUAD Team

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1 INTRODUCTION: THE KIDNEY

The kidneys are paired, bean-shaped organs that sit in the retroperitoneal space of the posterior abdominal wall. They are the principal organs of the urinary system and are responsible for maintaining homeostasis by filtering blood and producing urine.

Key Anatomical Facts

Position	Retroperitoneal (behind peritoneum)
Vertebral level	T11 – L3
Normal adult size	11 – 15 cm in length
Asymmetry	Right kidney is usually slightly shorter than the left (upper limit of variation = 1.5 cm)
Hilum	Medial depression where renal artery, vein, and ureter enter/exit

Gross Internal Structure

The cut surface of the kidney reveals two distinct zones. The outer **cortex** is the reddish-brown granular region that contains glomeruli, PCT, DCT, and parts of the collecting ducts. The inner **medulla** is organised into 8–18 cone-shaped **renal pyramids**, whose tips (papillae) project into cup-like minor calyces.

Minor calyces merge into major calyces, which converge to form the **renal pelvis** – a funnel that narrows into the ureter. The **renal sinus** is the fat-filled cavity at the hilum that holds the collecting system and vessels.

Layers Surrounding the Kidney

Renal capsule	Innermost fibrous layer; protects against infection/trauma
Adipose capsule	Middle cushion of fat around the capsule
Renal fascia	Outermost connective-tissue layer anchoring the kidney

2 FUNCTIONS OF THE KIDNEY

1. Excretory Function

The kidney excretes metabolic waste products, drugs, and toxins from the blood. This is the most widely recognised function and forms the basis of renal function tests.

2. Homeostatic Function

The kidney maintains the constancy of the internal environment through three key mechanisms:

- **Water balance** – adjusts urinary water loss to match intake
- **Electrolyte balance** – controls Na^+ , K^+ , Ca^{2+} , phosphate, and other ions
- **Acid-base balance** – secretes H^+ and reabsorbs HCO_3^- to regulate blood pH

3. Endocrine (Secretory) Function

Renin	Secreted by juxtaglomerular (JG) cells; activates the renin-angiotensin-aldosterone system (RAAS)
Erythropoietin (EPO)	Secreted by endothelial cells of peritubular capillaries in the renal cortex; stimulates red blood cell production
Prostaglandins	Local mediators of renal blood flow and salt/water excretion

4. Endocrine (Metabolic) Function

The kidney converts **Vitamin D₃** (cholecalciferol) into its active form, **1,25-dihydroxycholecalciferol (calcitriol)**, by the action of 1-alpha-hydroxylase enzyme in the cells of the PCT, under the influence of parathyroid hormone (PTH). Calcitriol promotes intestinal calcium absorption.

3 THE NEPHRON – FUNCTIONAL UNIT

The **nephron** is the structural and functional unit of the kidney. Each kidney contains approximately **1.2 million nephrons**. A nephron consists of two parts:

- **Renal corpuscle** – filters blood plasma to produce a filtrate
- **Long renal tubule** – processes the filtrate into urine

Flow of Glomerular Filtrate

Glomerular capsule (Bowman's space) → Proximal Convoluted Tubule (PCT) → Nephron loop (Loop of Henle) → Distal Convoluted Tubule (DCT) → Collecting Duct (CD) → Papillary duct → Minor calyx → Major calyx → Renal pelvis → Ureter → Urinary bladder → Urethra

4 RENAL CORPUSCLE (RENAL MALPIGHIAN CORPUSCLE)

The renal corpuscle (also called the Malpighian corpuscle) consists of the **glomerular (Bowman's) capsule** and the enclosed **glomerulus**.

Components

Glomerular capsule	Double-walled cup surrounding the glomerulus; outer wall = parietal layer (simple squamous epithelium); inner wall = visceral layer made of podocytes
Glomerulus	Tuft of fenestrated capillaries; high-pressure filtration bed
Capsular space	The space between the two walls; contains glomerular filtrate
Mesangial cells	Support cells between capillaries; can contract to modulate GFR

Two Poles of the Renal Corpuscle

- **Vascular pole** – where the **afferent arteriole enters** and the **efferent arteriole exits**
- **Urinary pole** – the parietal wall becomes continuous with the PCT on the opposite side

The Filtration Barrier (3 Layers)

For filtrate to enter the capsular space, fluid must pass through three barriers:

1. Fenestrated endothelium	Honeycomb-like capillary wall with pores of 70–90 nm; highly permeable but blocks blood cells
2. Basement membrane	Proteoglycan gel with a negative charge; excludes particles >8 nm; also repels negatively charged proteins such as albumin
3. Filtration slits	Formed by podocyte foot processes; allow particles <3 nm to pass freely; negatively charged (restricts anionic proteins)

What Passes Freely vs What is Retained

Passes freely (<3 nm)	Water, electrolytes, glucose, amino acids, fatty acids, nitrogenous wastes, vitamins
Retained in blood	Plasma proteins (>8 nm) and substances bound to them (iron, calcium, thyroid hormone)
Clinical note	Trauma, kidney infection, or strenuous exercise can damage the filtration barriers, causing proteinuria (albuminuria) and hematuria

5 RENAL TUBULE

The renal tubule leads away from the glomerular capsule and ends at the tip of the medullary pyramid. It consists of four major regions.

5.1 Proximal Convoluted Tubule (PCT)

- Arises directly from the glomerular capsule
- **Longest and most coiled** of the four regions
- Lined by **simple cuboidal epithelium with prominent microvilli** (brush border) – greatly increases absorptive surface area
- Rich in **mitochondria** to supply ATP for active transport
- **Reabsorbs ~65% of the glomerular filtrate**

5.2 Nephron Loop (Loop of Henle)

A U-shaped structure following the PCT that dips from the cortex into the medulla and returns. Its primary role is to generate a salinity (osmotic) gradient in the medullary interstitium that allows the collecting duct to concentrate urine.

Descending limb	Passes from cortex to medulla; thin segment lined by simple squamous epithelium; very permeable to water, low metabolic activity
Ascending limb	Thick segment (simple cuboidal epi.) is impermeable to water but actively transports Na^+ , K^+ , Cl^- outward; highly metabolically active

5.3 Distal Convoluted Tubule (DCT)

When the nephron loop returns to the cortex it becomes coiled again as the DCT. It is lined by **cuboidal epithelium without microvilli** (unlike PCT). The DCT marks the end of the nephron itself.

5.4 Collecting Duct (CD)

Several DCTs from multiple nephrons drain into a single straight collecting duct that passes through the medulla. Collecting ducts merge to form **papillary ducts** which drain into the minor calyx.

Two cell types in the DCT and CD:

Principal cells	Have receptors for aldosterone and ADH; regulate salt and water balance; secrete K^+
Intercalated cells	High in mitochondria; regulate acid-base balance; reabsorb K^+ and secrete H^+

6 TYPES OF NEPHRON

Cortical nephrons (85–90%)	Glomeruli located in the outer cortex; short loops of Henle that do NOT extend deeply into the medulla; supplied by peritubular capillaries
Juxtamedullary nephrons (10–15%)	Glomeruli located near the corticomedullary junction; long loops of Henle that extend deep into the medulla; supplied by the vasa recta (straight capillaries parallel to the loop); essential for producing concentrated urine

7 RENAL CIRCULATION & BLOOD FLOW

The kidneys account for only **0.4% of body weight** but receive approximately **25% (1300 mL/min) of the cardiac output**. This high blood flow is not for metabolic needs (kidneys use only 8% of total body O₂) but rather to support the high rate of glomerular filtration essential for homeostasis.

Arterial Supply Pathway

Renal artery → Segmental arteries → Interlobar arteries → Arcuate arteries → Interlobular arteries → Afferent arterioles → Glomerulus → Efferent arteriole → Peritubular capillaries (cortical) / Vasa recta (juxtamedullary) → Interlobular veins → Arcuate veins → Interlobar veins → Renal vein

Key Feature

Blood passes through **two capillary beds in series** – the glomerular capillaries (high pressure, for filtration) and the peritubular/vasa recta capillaries (low pressure, for reabsorption). This unique double-capillary arrangement is possible because of the intervening efferent arteriole.

Innervation

Sympathetic nerve fibres from the superior mesenteric ganglion enter the kidney at the hilum, following the renal artery to innervate the afferent and efferent arterioles. Their main roles are:

- Regulate blood flow and GFR via vasoconstriction/dilation
- Stimulate renin secretion when blood pressure falls

8 URINE FORMATION: OVERVIEW

Urine formation involves three integrated processes that transform blood plasma into the final urine product.

1	Glomerular Filtration	Renal corpuscle	Plasma filtrate enters capsular space
2	Tubular Reabsorption	PCT, Loop, DCT, CD	Useful solutes and water return to blood
3	Tubular Secretion	PCT, Loop, DCT	Wastes and excess ions added to filtrate

Of the **180 L/day** of filtrate produced, **99%** is reabsorbed, leaving only **1–2 L of urine** per day.

9 STAGE 1: GLOMERULAR FILTRATION

Filtration Pressure

Glomerular filtration is driven by a net outward pressure, the **Net Filtration Pressure (NFP)**:

$$\text{NFP} = \text{Blood Hydrostatic Pressure (BHP)} - \text{Capsular Pressure (CP)} - \text{Colloid Osmotic Pressure (COP)}$$

$$\text{NFP} = 60 \text{ mmHg}_{\text{out}} - 18 \text{ mmHg}_{\text{in}} - 32 \text{ mmHg}_{\text{in}} = \mathbf{10 \text{ mmHg outward}}$$

BHP (60 mmHg, outward)	High blood pressure in glomerular capillaries; kept high because afferent arteriole is wider than efferent
CP (18 mmHg, inward)	Hydrostatic pressure of fluid already in capsular space opposing filtration
COP (28–32 mmHg, inward)	Colloid osmotic pressure due mainly to albumin in plasma; opposes filtration (same as elsewhere in body)

Glomerular Filtration Rate (GFR)

GFR is the total volume of filtrate formed per minute by both kidneys combined. It is expressed as: **GFR = NFP × K**

where K (filtration coefficient) = 12.5 mL/min per mmHg (depends on permeability and surface area of the filtration barrier).

GFR – Males	~125 mL/min or ~180 L/day
GFR – Females	~105 mL/min or ~150 L/day

10 FACTORS AFFECTING & REGULATION OF GFR

Factors That Affect GFR

1. Filtration coefficient (K)	Increased permeability (hypoxia, toxins) raises GFR; contraction of mesangial cells reduces filtration surface and decreases GFR
2. Capsular (Bowman's) pressure	Acute urinary obstruction raises capsular pressure → decreases GFR
3. Blood hydrostatic pressure	GFR is stable between MAP 80–200 mmHg; falls in acute renal failure
4. Renal blood flow	Altered by arteriole resistance; afferent/efferent dilation or constriction changes GFR
5. Blood oncotic pressure	Hyperproteinaemia → increased COP → decreased GFR
6. Sympathetic stimulation	Minor direct effect; decreases GFR
7. State of glomerular membrane	Disease can alter permeability or surface area

Regulation of GFR – Why It Must Be Controlled

- Too HIGH GFR → excessive urine → **dehydration and electrolyte depletion**
- Too LOW GFR → wastes reabsorbed → **azotemia (waste build-up in blood)**

GFR is controlled by adjusting glomerular blood pressure via three homeostatic mechanisms:

Mechanism 1: Renal Autoregulation

The ability of the nephrons to adjust blood flow and GFR **without external control**. Maintains stable GFR over a MAP range of **80–170 mmHg**.

Two sub-mechanisms:

- **Myogenic mechanism:** Rising BP stretches the afferent arteriole smooth muscle → it contracts → reduces glomerular blood flow → GFR returns to normal. Conversely, falling BP relaxes smooth muscle → vasodilation → GFR is restored.
- **Tubuloglomerular feedback:** Macula densa cells (specialised cells in DCT at the JGA) monitor tubular fluid NaCl. If GFR rises, more NaCl reaches the macula densa → it signals juxtaglomerular cells to constrict the afferent arteriole → reduces GFR.

Mechanism 2: Sympathetic Control of GFR

During strenuous exercise or circulatory shock, SNS activation and adrenal epinephrine cause afferent arteriole constriction, redirecting blood to heart, brain, and skeletal muscles. Result: reduced GFR and urine production.

Mechanism 3: Hormonal Control – Renin-Angiotensin System

When BP drops, SNS stimulates juxtaglomerular cells to secrete **renin** (enzyme). Renin converts angiotensinogen (liver protein) to **Angiotensin I (10 aa)**. ACE (angiotensin-converting enzyme) in the lungs then converts it to **Angiotensin II (8 aa)**.

Angiotensin II acts to raise BP by:

- Causing vasoconstriction of blood vessels

- Stimulating the adrenal cortex to secrete aldosterone
- Stimulating thirst via the hypothalamus

Aldosterone then acts on the DCT/CD to promote Na^+ reabsorption → water reabsorption → increased blood volume → restored BP.

Summary: Factors Affecting GFR via Mesangial Cells

Contraction (decrease GFR)	Relaxation (increase GFR)
Angiotensin I	Atrial natriuretic peptide (ANP)
Antidiuretic hormone (ADH)	Cyclic AMP
Endothelins	Dopamine
Histamine	Prostaglandins (PGE_2)
Leukotrienes C_4 , D_4	Nitric oxide (NO)
Prostaglandins (PGF_2)	
Platelet-derived growth factor (PDGF)	
Platelet activating factor	
Thromboxane- A_2	

11 STAGE 2: TUBULAR REABSORPTION

Tubular reabsorption is the process of reclaiming water and useful solutes from the tubular fluid and returning them to the blood via the peritubular capillaries. It is the reverse of filtration.

Two Routes of Reabsorption

Transcellular route	Substances pass through the cytoplasm of epithelial cells and out through their basolateral surface
Paracellular route	Substances pass between epithelial cells through tight junctions and enter the ECF, then the peritubular capillaries

Reabsorption in the PCT

65% of the glomerular filtrate is reabsorbed here. **Sodium (Na^+)** is the key driver – its reabsorption creates an osmotic and electrical gradient that drives reabsorption of water and other solutes.

- **Water:** 2/3 of filtered water reabsorbed by osmosis
- **Na^+ , Cl^- :** active and passive transport
- **Glucose, amino acids:** completely reabsorbed via sodium co-transport (SGLT)
- **HCO_3^- :** reabsorbed via Na^+ -H exchange and carbonic anhydrase
- Reabsorption of Na^+ and other solutes makes tubular cells **hypertonic** → water follows by osmosis

Reabsorption in the Nephron Loop

The primary role of the loop is to create the medullary osmotic gradient:

- **Descending limb:** Permeable to water (4+); water exits by osmosis as the fluid flows into the increasingly salty medulla; fluid becomes more concentrated
- **Ascending limb:** Impermeable to water (0); actively transports $\text{Na}^+/\text{K}^+/\text{2Cl}^-$ (NaKCC) outward; fluid becomes dilute (~150 mOsm/L) as it re-enters cortex

Reabsorption in the DCT and Collecting Duct

Fluid arriving at the DCT is about 20% water and 7% salts of the original filtrate. DCT and CD are hormonally regulated:

Aldosterone	$\uparrow \text{Na}^+$ reabsorption → $\uparrow$ water reabsorption → $\downarrow$ urine volume; triggered by angiotensin II
ADH (vasopressin)	Secreted by posterior pituitary in response to dehydration or $\uparrow$ blood osmolarity; inserts aquaporin-2 channels in CD → makes CD more permeable to water → water reabsorbed → concentrated urine
ANP (Atrial Natriuretic Peptide)	Released by atria when BP $\uparrow$; dilates afferent arteriole, inhibits RAAS and ADH, inhibits NaCl reabsorption → $\uparrow$ excretion of salt and water → $\downarrow$ BP
PTH (Parathyroid Hormone)	$\uparrow \text{Ca}^{2+}$ reabsorption in DCT; $\uparrow$ phosphate excretion in PCT; stimulates calcitriol production

Obligatory vs. Facultative Water Reabsorption

Obligatory reabsorption (~85%)	Always occurs irrespective of body water balance; by osmosis in response to solute reabsorption in the PCT and loop
Facultative reabsorption (~15–18%)	Occurs in the collecting duct under ADH control; depends on hydration state

12 STAGE 3: TUBULAR SECRETION

Tubular secretion is the process by which the renal tubule extracts chemicals (wastes and excess ions) from the peritubular capillary blood and secretes them into the tubular fluid for excretion. It has two main purposes:

1. Waste removal	Secretes urea, uric acid, bile acids, ammonia, catecholamines, prostaglandins, creatinine, penicillin, morphine, aspirin, and other drugs. This is why many drugs must be taken 3–4x daily – the kidney clears them quickly.
2. Acid-base balance	Secretion of H^+ and reabsorption/generation of HCO_3^- maintains blood pH. H^+ secreted in the PCT is used for HCO_3^- reabsorption; H^+ secreted by intercalated cells in the DCT/CD maintains urinary acidity.

Secreted Substances (Proximal Tubule)

PAH, cAMP, bile salts, oxalate, prostaglandins, water	Acetazolamide, furosemide, penicillin, probenecid, salicylate (aspirin), NSAIDs
Creatinine, dopamine, epinephrine, norepinephrine	Atropine, cimetidine, morphine, quinine, procainamide, verapamil

13 KIDNEY HANDLING OF SPECIFIC SUBSTANCES

13.1 Water

Of the **180 L/day** filtered, only 1–2 L are excreted. About **2/3 is reabsorbed in the PCT**, 15% in the loop, and the remaining under hormonal (ADH) control in the DCT/CD.

13.2 Sodium (Na⁺)

Na⁺ is the master solute driving reabsorption. It is the most abundant cation in the filtrate. It is reabsorbed by both transcellular and paracellular routes throughout the nephron. Its reabsorption drives obligatory water reabsorption.

13.3 Potassium (K⁺)

Freely filtered; almost completely reabsorbed in the PCT and loop (so little reaches DCT). K⁺ secretion in the DCT/CD is regulated by:

- Plasma K⁺ level – hyperkalemia stimulates secretion
- Aldosterone – promotes K⁺ secretion via 3 mechanisms: (a) increases Na⁺-K⁺ pump activity; (b) increases Na⁺ absorption creating negative lumen; (c) alters apical membrane K⁺ channel permeability
- GFR and tubular blood flow – higher flow = more K⁺ secretion
- ADH – dual, balanced effect
- Acidity – inhibits K⁺ secretion

13.4 Glucose

Glucose is completely reabsorbed in the PCT under normal blood glucose levels via the **SGLT** (sodium-dependent glucose transporter) on the apical membrane and **GLUT1/2** on the basolateral membrane.

Renal threshold	180 mg/dL – at this plasma glucose level, carriers begin to saturate
Transport maximum (T_{mG})	375 mg/min – beyond this, glucose appears in urine (glycosuria)
Splay	The gradual rather than abrupt appearance of glucose in urine due to heterogeneity of nephron T _{mG} values

*At 300 mg/dL plasma glucose, all carriers are saturated and glucose spills into urine. **Diabetes mellitus** causes glycosuria because hyperglycaemia exceeds the renal threshold.*

13.5 Urea

Urea is a major nitrogenous waste. At a plasma flow of 750 mL/min, urea clearance is about 75 mL/min (10%). Urea amount filtered depends on renal blood flow and protein intake. Proximal straight tubules and thin descending/ascending segments secrete urea into tubular fluid. The inner medullary collecting duct reabsorbs urea via the UT-A transporter (stimulated by ADH) – this increases medullary urea concentration, contributing to the osmotic gradient.

13.6 Uric Acid

Freely filtered; undergoes 4-stage handling in the PCT: 90% reabsorbed at S1, 50% secreted at S2, then 40% post-secretory reabsorption at S3, leaving ~10% excreted. Transporters: urate transport protein (apical), anion exchanger (basolateral).

13.7 Para-Amino Hippuric Acid (PAH)

PAH is an exogenous organic acid used to measure renal plasma flow. 90% of PAH is bound to plasma proteins, so it is excreted by both glomerular filtration and tubular secretion (in proximal tubules). PAH characteristics for tubular secretion: carrier-mediated, non-selective, saturable, occurs in proximal tubules.

13.8 Acid-Base Balance

H^+ secreted in the PCT (in exchange for Na^+) is used to regenerate HCO_3^- . H^+ combines with filtered HCO_3^- in the lumen, forming H_2CO_3 , which dissociates into $CO_2 + H_2O$ (catalysed by carbonic anhydrase). CO_2 re-enters the tubular cell, regenerates H^+ and HCO_3^- intracellularly; HCO_3^- exits to blood via $Na-HCO_3^-$ symporter.

Regulation of HCO_3^- reabsorption depends on:

1. Plasma HCO_3^- level
2. Plasma PCO_2 level
3. Extracellular volume ($\uparrow$ ECF $\rightarrow$ $\downarrow$ HCO_3^- secretion)
4. Aldosterone and angiotensin II (elevated $\rightarrow$ $\uparrow$ HCO_3^- reabsorption)
5. PTH inhibits HCO_3^- absorption
6. Plasma K^+ : hyperkalemia inhibits; hypokalemia stimulates H^+ secretion

ADH – Summary of Functions

- Increases Na^+ and water absorption (via aquaporins) – affects tubular fluid flow $\rightarrow$ leads to K^+ secretion
- Makes tubular lumen more negative $\rightarrow$ favours K^+ secretion
- Affects urea transport protein $\rightarrow$ increases urea absorption $\rightarrow$ increases water absorption

14 WATER CONSERVATION & CONCENTRATION OF URINE

The kidneys produce urine of variable concentration depending on the body's hydration state. The key is the medullary osmotic gradient.

Control of Water Loss

Overhydration (↑ fluid intake)	Blood osmolarity falls → no ADH released → CD stays impermeable to water → water remains in tubular fluid → dilute (hypotonic) urine = water diuresis
Dehydration	Blood osmolarity rises → hypothalamus stimulates posterior pituitary → ADH released → aquaporin channels inserted in CD → CD more permeable to water → water reabsorbed → concentrated (hypertonic) urine

The Collecting Duct in Detail

Urine entering the upper collecting duct is isotonic (~300 mOsm/L). Two conditions allow it to become hypertonic by the time it leaves:

- The ECF osmolarity in the lower medulla is 4x higher than in the cortex (up to 1200 mOsm/L)
- The medullary portion of the CD is more permeable to water than to NaCl

As urine flows down the CD through the increasingly salty medulla, water leaves by osmosis down the concentration gradient, making the urine progressively more concentrated. Final urine osmolarity can reach 1200 mOsm/L.

Permeability of Nephron Segments (Summary Table)

Proximal tubule	++	+	+	2+
Thin Descending Limb	4+	+	±	0
Thin Ascending Limb	0	+	4+	0
Thick Ascending Limb	0	±	±	4+
Distal Convoluted Tubule	±	±	±	3+
Cortical Collecting Tubule	3+	0	±	2+
Outer Medullary Collecting Tubule	3+	0	±	1+
Inner Medullary Collecting Tubule	3+	3+	±	1+

*Note: The thick ascending limb of the loop is **impermeable to water (0)** – this is the key driver of urine dilution.*

15 COUNTERCURRENT MECHANISM

Countercurrent Multiplication

Is the process by which a progressively increasing osmotic gradient (from 300 to 1200 mOsm/L) is generated in the **interstitial fluid of the renal medulla** as a result of countercurrent (opposite direction) flow in the two limbs of the loop of Henle.

Steps in generating the gradient:

1. The thick ascending limb actively transports NaCl out but is impermeable to water → makes medullary interstitium hypertonic
2. The descending limb is highly permeable to water → water leaves by osmosis → fluid in the descending limb becomes more concentrated toward the tip
3. As more NaCl is pumped out by the ascending limb and more water leaves the descending limb, the osmotic gradient is 'multiplied' – progressively increasing toward the papilla

Countercurrent Exchange (Vasa Recta)

The vasa recta (straight capillaries alongside juxtamedullary nephron loops) passively exchange solutes and water with the medullary interstitium via countercurrent flow. This prevents washout of the medullary gradient while still supplying nutrients to medullary cells.

Formation of Dilute Urine – When ADH is Absent

When ADH is absent, the principal cells of the late DCT and collecting duct are impermeable to water. Solutes (NaCl) are reabsorbed but water remains in the tubule. The final urine osmolarity can be as low as 50–65 mOsm/L. Tubular fluid entering the early distal tubule is always hypotonic (~100–150 mOsm/L) because solutes are pumped out of the thick ascending limb without water.

16 COMPOSITION & PROPERTIES OF URINE

Physical Properties

Volume	Normal: 1–2 L/day; Output <400 mL/day → azotemia
Colour	Pale yellow (dilute) to amber (concentrated); due to urobilin
Transparency	Clear when fresh; may become cloudy on standing
Odour	Fresh urine has distinct odour; bacteria degrade urea → ammonia (pungent); DM → fruity/acetone odour; UTI → rotten odour
pH	Range 4.5–8.2; usually 6.0 (mildly acidic)
Specific gravity	1.001–1.028
Osmolarity	50–1200 mOsm/L (blood = 300 mOsm/L)

Chemical Composition

95% water and 5% solutes. Normal solutes include:

Urea	Most abundant organic solute; 1.8 g/dL; 21 g/day
Creatinine	150 mg/dL; 1.8 g/day – used to estimate GFR
Uric acid	40 mg/dL; 0.5 g/day
Chloride	533 mg/dL; 6.4 g/day
Sodium	333 mg/dL; 4 g/day
Potassium	166 mg/dL; 2 g/day
Ammonia	60 mg/dL; 0.68 g/day

*It is **abnormal** to find glucose, free haemoglobin, albumin, ketones, or bile pigments in the urine – their presence indicates disease.*

Urine Volume Terminology

Normal	1–2 L/day
Polyuria	>2 L/day (high fluid intake, diuretics, DM)
Oliguria	<500 mL/day (dehydration, kidney disease, shock, obstruction)
Anuria	0–100 mL/day (severe renal failure, complete obstruction)

17 CLINICAL CORRELATES

17.1 Diabetes Mellitus (DM) & Glycosuria

In DM, persistently high plasma glucose exceeds the renal threshold (180 mg/dL), saturating SGLT transporters. Glucose then appears in the urine (glycosuria). The osmotic effect of glucose in the tubular fluid also causes osmotic diuresis (polyuria), resulting in polydipsia.

17.2 Diabetes Insipidus (DI)

Results from deficiency of ADH (central DI) or resistance to ADH (nephrogenic DI). Without ADH, the CD cannot reabsorb water → massive polyuria (up to 15–20 L/day) of very dilute urine. Unlike DM, urine does not contain glucose (not sweet).

17.3 Proteinuria & Haematuria

Damage to the filtration barrier (e.g. from trauma, infection, strenuous exercise, glomerulonephritis) allows proteins (albuminuria) and red blood cells (haematuria) to pass into the filtrate and appear in the urine.

17.4 Hypokalemia → Metabolic Alkalosis

In hypokalemia, K^+ moves out of cells into the ECF, and H^+ moves into cells to maintain electroneutrality. This raises plasma HCO_3^- and lowers intracellular pH. The intracellular acidosis in renal tubular cells promotes H^+ secretion and HCO_3^- reabsorption → metabolic alkalosis.

17.5 Diuretics

Chemicals that increase urine volume by either:

- Increasing GFR (e.g. caffeine dilates afferent arteriole)
- Decreasing tubular reabsorption (e.g. alcohol inhibits ADH secretion)

Used clinically in hypertension and congestive heart failure.

17.6 Azotemia

When urine output drops below 400 mL/day, the kidneys cannot maintain safe, low concentrations of waste products (urea, creatinine) in the plasma. This build-up of nitrogenous wastes is called azotemia and represents a critical indicator of renal failure.

17.7 Hemodialysis

In end-stage renal disease, when the kidneys can no longer adequately filter blood, hemodialysis provides an artificial substitute. Blood is drawn from a surgically created arteriovenous shunt in the arm, pumped through a dialysis machine where it passes next to a semipermeable membrane bathed in dialysis fluid, allowing diffusion of waste products down their concentration gradients, and returned to the body.